

PRN:

Batch:

Experiment no 2

Title: Parallel Region Creation and Thread Identification using OpenMP

Tasks:

1. Write an OpenMP "Hello World" program to create a parallel region and print a message from each thread.
2. Display the thread ID and total number of threads using `omp_get_thread_num()` and `omp_get_num_threads()`.
3. Execute the program using different numbers of threads (1, 2, 4, 8, and maximum available threads) and observe the output.
4. Retrieve the following runtime information:
 - Number of processors (`omp_get_num_procs()`)
 - Maximum number of threads (`omp_get_max_threads()`)
 - Dynamic thread status
 - Nested parallelism status
5. Write a simple computational program for summation of two large arrays of size 10,000 and execute it with different thread counts.
6. Measure the execution time using `omp_get_wtime()` for each thread configuration.
7. Compute the following performance metrics:

- Speedup

$$S = \frac{T_{serial}}{T_{parallel}}$$

- Efficiency

$$E = \frac{S}{P} \times 100\%$$

where P is the number of threads.

8. Compare the observed speedup with the theoretical expectation based on **Amdahl's Law**, and discuss why linear speedup is not always achieved.
9. Plot the following graphs:
 - Number of Threads vs Execution Time
 - Number of Threads vs Speedup
 - Number of Threads vs Efficiency
10. Summarize the observations and discuss the effect of increasing the number of threads on performance.